

Financial Flows

General rule

No money movement may bypass the Ledger. Wallet current balance is operational state; Ledger and FinancialTransaction records provide the explainable financial history.

1. Wallet-to-Wallet Transfer — implemented

Endpoint: POST /api/v1/transfers.

Flow:

```
Gateway JWT
-> API JWT + durable session
-> completed idempotency replay
-> server-side risk signals
-> internal fraud
-> external FakeFraud via Gateway
-> final Allow
-> one MSSQL financial transaction
    -> idempotency
    -> source debit
    -> destination credit
    -> FinancialTransaction
    -> LedgerJournal + LedgerEntries
-> commit
```

Accounting:

```
Debit    Source Wallet Liability
Credit   Destination Wallet Liability
```

Replaying the same request/key never moves money a second time.

2. BankDeposit — planned next flow

The purpose is to fund a FinWallet wallet after an external bank movement has been confirmed. No public endpoint exists yet.

Proposed lifecycle:

```
Created
-> Fraud/Cutoff if required
-> BankPending
-> BankConfirmed
-> LedgerPost
-> Completed
```

The wallet must not be credited before external confirmation. Deposit posting should create a balanced journal between a provider-side asset/clearing account and the customer wallet liability.

3. BankWithdrawal — planned

Withdrawal is a long-running external workflow, so available funds should first move into blocked state.

Proposed state:

```
Available -> Blocked -> external bank pending
                -> success: settle
                -> failure: release/compensate
```

The SQL transaction must not remain open during external HTTP.

4. Refund / Reversal — planned public flow

Refund is a business-level return, while reversal is the accounting inverse of a previous financial effect. The original transaction/journal is not deleted or mutated. A new FinancialTransaction and reverse

journal are created.

5. Campaign / Merchant Purchase — planned

FakeCampaign may calculate eligibility and discount, but FinWallet Ledger remains responsible for correctly allocating the economic effect across customer, merchant and sponsor accounts.

6. Fraud Review

A Review decision currently moves no money. Durable FraudEvents/manual-review queueing is not implemented yet.

7. Cutoff

FakeCutoff evaluates processing/settlement eligibility by bank, currency and transaction type. This integration is mainly planned for BankDeposit and BankWithdrawal.

8. Notification

Notification failure after a financial commit must not reverse the money transaction. Reliable post-commit notification is therefore planned around an Outbox pattern.

9. Reconciliation

Planned checks include:

- wallet current state vs ledger-derived balance;
- internal bank-related transactions vs FakeBank statement;
- orphaned idempotency/transaction/ledger records.

Mismatches are recorded and investigated rather than silently fixed with an UPDATE.

10. Current demo limitation

New wallets start at zero balance. Because public BankDeposit does not yet exist, a new customer cannot currently be fully funded and transferred using only public endpoints. An integration fixture must atomically create balance + FinancialTransaction + balanced LedgerJournal/Entries; directly updating Wallets.AvailableBalance is invalid.